Supplementary Material Concerning the Paper

Synthesis of Optimal Stealthy Sensor Attackers Against Current-State Opacity

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In this supplementary material, we present the complete ACAG and AO-ACAG constructed for the running example in our paper, shown in Fig. 1 and Fig. 2, respectively. These graphs are automatically generated by our developed tool available in [30]. Note that, for the sake of implementation convenience, we write the state estimates ξ_{Si} and ξ_{Ai} as S_i and A_i, respectively, and their corresponding sets are detailed in Table I. We observe that S_i in Table I is a set of state pairs, which differs from its definition. These variations are adopted to facilitate the implementation while preserving the core logic of the proposed algorithms. Specifically, each pair in S_i consists of a supervisor state z and a plant state x. While the collection of all plant states (the second entries of the pairs) constitutes the supervisor estimate as defined, S_i explicitly associates each supervisor state with possible plant states. Consequently, S_0 is the same as S_3 and S_2 is the same as S_4 in Table I according to the formal definition of S_i in our paper. We note that, despite these additional details, the generated ACAG remains fully consistent with its definition.

TABLE I: State estimates of the attacker and the supervisor

Supervisor Estimates	**Attacker Estimates**
$S_0 : \{(0,\underline{5}),(0,\underline{0})\}$	$A_0 : \{3\}$
$S_1 : \{(0,\underline{0})\}$	$A_1 : \{5\}$
$S_2 : \{(0,\underline{2}),(0,\underline{1})\}$	$A_2 : \{6\}$
$S_3 : \{(1,\underline{5}),(1,\underline{0})\}$	$A_3 : \{8\}$
$S_4 : \{(1,\underline{1}),(1,\underline{2})\}$	$A_4 : \{4,6\}$
$S_5 : \{(1,\underline{6}),(1,\underline{7}),(1,\underline{3}),(1,\underline{4})\}$	$A_5 : \{1,2,5\}$
$S_6 : \{(1,\underline{6}),(1,\underline{7}),(1,\underline{4})\}$	$A_6 : \{4,6,7\}$
$S_7 : \{(1,\underline{7}),(1,\underline{4})\}$	$A_7 : \{0,1,2,5\}$
$S_8 : \{(2,\underline{3}),(2,\underline{6})\}$	-

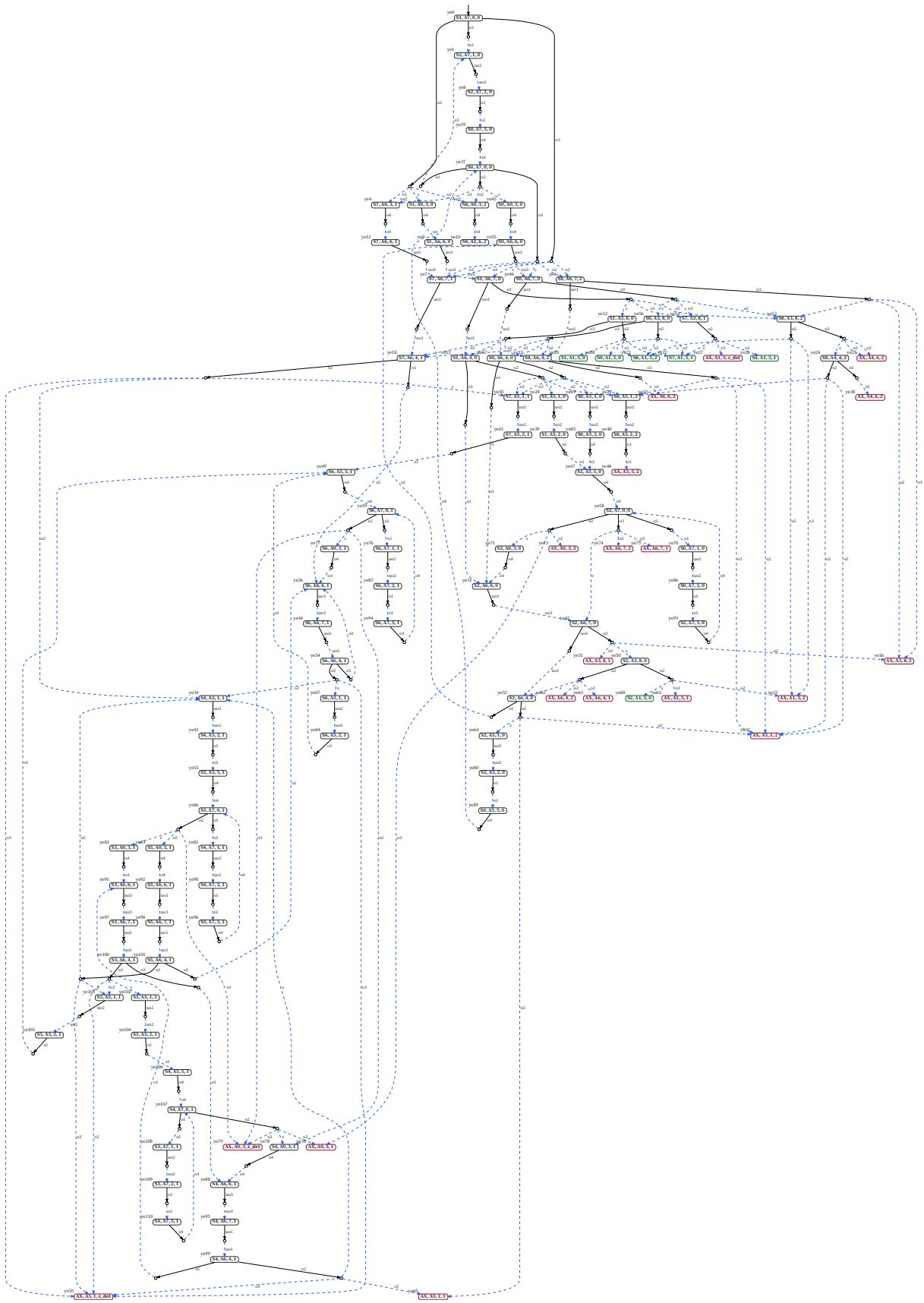

Fig. 1: ACAG $\mathcal{Y}$

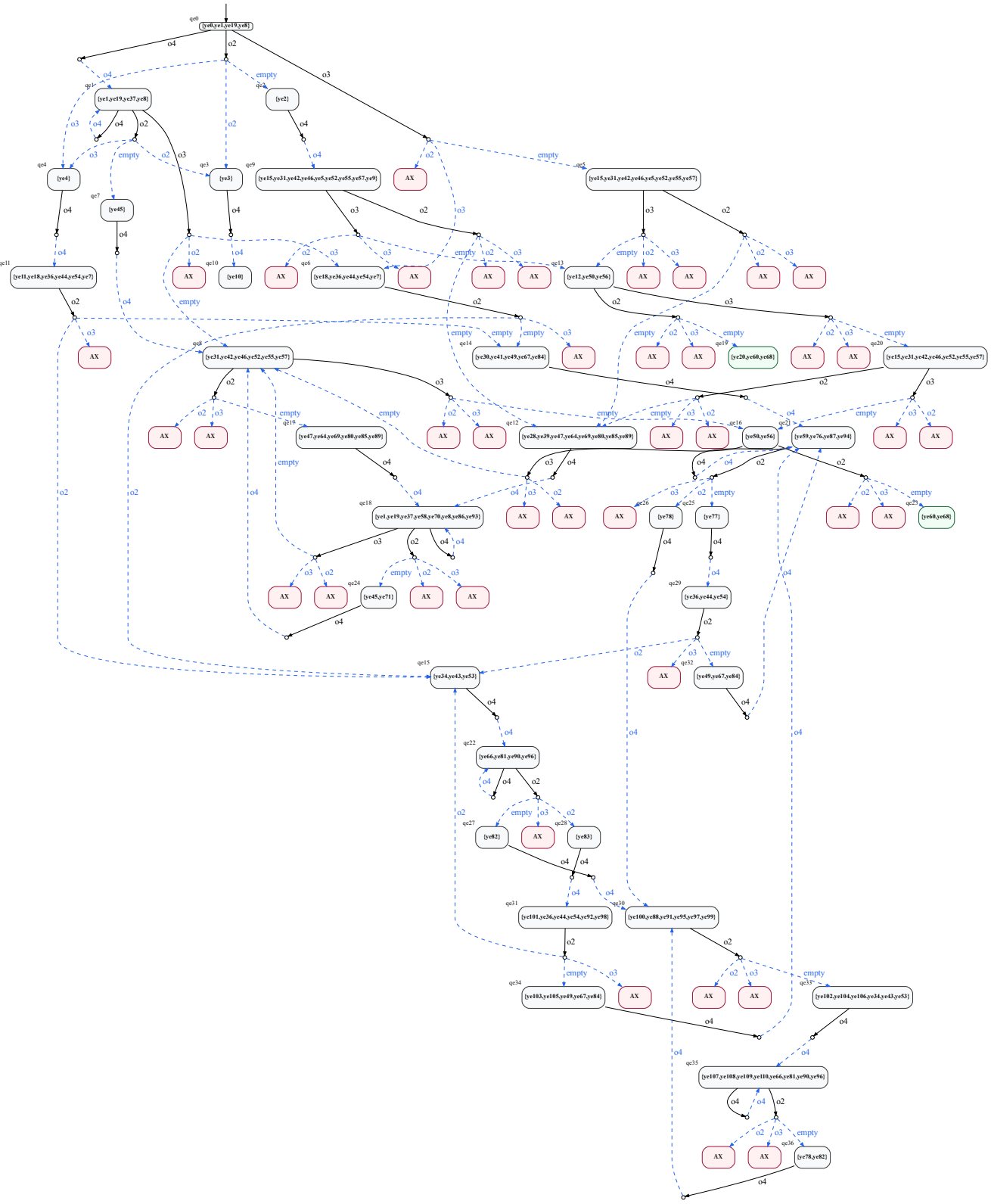

Fig. 2: AO-ACAG $\mathcal{Y}^Q$